

Foundations of Instruction tuning

[Instruction tuning]

1.0 Content Block

In late 2024, a new approach to LLM development emerged with "reasoning models". These are trained to generate step-by-step analysis before producing final answers, enabling better results on complex tasks, for instance in mathematics, coding and logic. OpenAI introduced this concept with their o1 model in September 2024, followed by o3 in April 2025. On the International Mathematics Olympiad qualifying exam problems, GPT-4o achieved 13% accuracy while o1 reached 83%. In January 2025, the Chinese company DeepSeek released DeepSeek-R1, a 671-billion-parameter open-weight reasoning model that achieved comparable performance to OpenAI's o1 while being significantly more cost-effective to operate. Unlike proprietary models from OpenAI, DeepSeek-R1's open-weight nature allowed researchers to study and build upon the algorithm, though its training data remained private. These reasoning models typically require more computational resources per query compared to traditional LLMs, as they perform more extensive processing to work through problems step by step.

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Stereotyping AI models can reinforce a wide range of stereotypes due to generalization, including those based on gender, ethnicity, age, nationality, religion, or occupation. When replacing human representatives, this can lead to outputs that homogenize or generalize groups of people. In 2023, LLMs assigned roles and characteristics based on traditional gender norms. For example, models might associate nurses or secretaries predominantly with women and engineers or CEOs with men due to the frequency of these associations in documented reality.

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$\log \blacksquare (\text{Perplexity}) = -\frac{1}{N} \sum_{i=1}^N \log \blacksquare (\Pr(\text{token}_i \blacksquare \text{context for token}_i))$ $\{\displaystyle \log(\text{Perplexity}) = -\frac{1}{N} \sum_{i=1}^N \log(\Pr(\text{token}_i | \text{context for token}_i))\}$

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Sycophancy Sycophancy is a tendency to agree with, flatter, or validate a user's stated beliefs rather than to prioritize factuality or corrective information. Continued sycophancy from LLMs has led to the observation of getting "1-shotted", denoting instances where conversational interaction with a large language model produces a lasting change in a user's beliefs or decisions, similar to the negative effects of psychedelics, and controlled experiments show that short LLM dialogues can generate measurable opinion and confidence shifts comparable to human interlocutors. Empirical analyses attribute part of the effect to human preference signals and preference models that reward convincingly written agreeable responses, and subsequent work has extended evaluation to multi-turn benchmarks and proposed interventions such as synthetic-data finetuning, adversarial evaluation, targeted preference-model reweighting, and multi-turn sycophancy benchmarks to measure persistence and regression risk. Industry responses have combined research interventions with product controls, for example Google and other labs publishing synthetic-data and fine-tuning interventions and OpenAI rolling back an overly agreeable GPT-4o update while publicly describing changes to feedback collection, personalization controls, and evaluation procedures to reduce regression risk and improve long-term alignment with user-level safety objectives. Mainstream culture has reflected anxieties about this dynamic where South Park satirized overreliance on ChatGPT and the tendency of assistants to flatter user beliefs in Season 27 episode "Sickofancy", and continued the themes across the following season, which commentators interpreted as a

critique of tech sycophancy and uncritical human trust in AI systems.

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demonstrates how to increase efficient exercise work by running up and down balls. moves all his arms and legs and builds up a lot of muscle. then plays the ball and we see a graphics and hedge trimming demonstration. performs sit ups while on the ball and talking.

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Tool use Tool use is a mechanism that enables LLMs to interact with external systems, applications, or data sources. It can allow for example to fetch real-time information from an API or to execute code. A program separate from the LLM watches the output stream of the LLM for a special tool-calling syntax. When these special tokens appear, the program calls the tool accordingly and feeds its output back into the LLM's input stream. Early tool-using LLMs were fine-tuned on the use of specific tools. But fine-tuning LLMs for the ability to read API documentation and call APIs correctly has greatly expanded the range of tools accessible to an LLM.

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Tokenization As machine learning algorithms process numbers rather than text, the text must be converted to numbers. In the first step, a vocabulary is decided upon, then integer indices are arbitrarily but uniquely assigned to each vocabulary entry, and finally, an embedding is associated with the integer index. Algorithms include byte-pair encoding (BPE) and WordPiece. There are also special tokens serving as control characters, such as [MASK] for masked-out token (as used in BERT), and [UNK] ("unknown") for characters not appearing in the vocabulary. Also, some special symbols are used to denote special text formatting. For example, "▮" denotes a preceding whitespace in RoBERTa and GPT and "##" denotes continuation of a preceding word in BERT. For example, the BPE tokenizer used by the legacy version of GPT-3 would split tokenizer: texts -> series of numerical "tokens" as

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In 2020, OpenAI researchers demonstrated that their new model GPT-3 could understand what format to use given a few rounds of Q and A (or other type of task) in the input data as example, thanks in part due to the RLHF technique. This technique, called few-shot prompting, allows LLMs to be adapted to any task without requiring fine-tuning. Also in 2022, it was found that the base GPT-3 model can generate an instruction based on user input. The generated instruction along with user input is then used as input to another instance of the model under a "Instruction: [...], Input: [...], Output:" format. The other instance is able to complete the output and often produces the correct answer in doing so. The ability to "self-instruct" makes LLMs able to bootstrap themselves toward a correct answer.

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Performance of bigger models on various tasks, when plotted on a log-log scale, appears as a linear extrapolation of performance achieved by smaller models. However, this linearity may be punctuated by "break(s)" in the scaling law, where the slope of the line changes abruptly, and where larger models acquire "emergent abilities". They arise from the complex interaction of the model's components and are not explicitly programmed or designed. One of the emergent abilities is in-context learning from example demonstrations. In-context learning is involved in tasks, such as: